PLAID: Training-Free Layout-Controllable Pattern Generation

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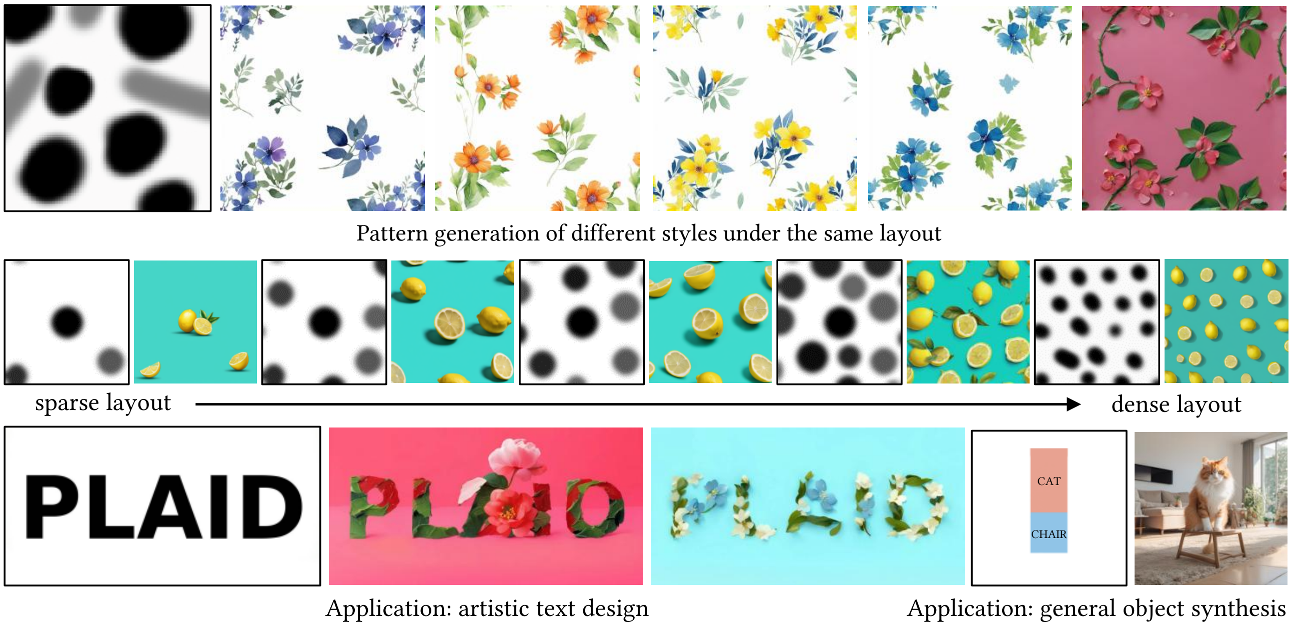

Figure 1: We introduce PLAID, a training-free framework for layout-controllable pattern generation. PLAID allows users to control layout using simple interactions based on randomly generated or hand-drawn black dots to specify the foreground element regions and grey dots to specify the flexible transition regions.

Abstract

Text-to-image diffusion models achieve impressive visual quality, but controllable pattern generation remains challenging because text prompts alone cannot precisely specify the layout of repeated patterns. Existing controllable generation methods either require extra training or are poorly suited to repeated pattern synthesis, often causing artifacts such as rigid structures, semantic leakage, and cluttered backgrounds. To address this problem, we propose a training-free framework for layout-controllable pattern generation, which introduces a flexible interaction paradigm, enabling users to specify layouts using extremely simple hand-drawn dots or adjustable density parameters. Specifically, our method performs attention-level semantic decoupling by keeping the spatial query shared while splitting the text-conditioned key-value branches for foreground and background. In addition, we introduce a background purification strategy based on an initial distribution shift, effectively suppressing unintended structural hallucinations. Extensive experiments show that our method achieves the best overall trade-off among semantic fidelity, layout adherence, visual quality, and efficiency in controllable pattern generation.

CCS Concepts

• **Computing methodologies** → **Image manipulation**; *Computer vision*; *Image processing*.

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Keywords

Pattern Design; Layout Control; Attention; Multi-Branch; Image Generation

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1 Introduction

The rapid advancement of diffusion models [7, 16, 18, 21] has revolutionized visual content creation, enabling the synthesis of high-quality images from textual descriptions. Within this domain, **pattern generation** plays an important role in computer graphics and vision. Unlike general object generation, pattern generation requires the harmonious repetition and arrangement of visual motifs and holds significant potential for applications in fashion design, digital art, and procedural game asset creation. As these applications increasingly demand not only aesthetic quality but also flexible user control over the generated designs, the creation of intelligent and flexible generation frameworks becomes essential.

Pattern generation is distinct from general scene generation. The latter typically focuses on compositional scenes composed of distinct semantic entities, such as an "apple on a table" or a "cat in a living room". In contrast, pattern generation often requires producing a large number of repeated instances with the same or similar semantics within a single image, such as "multiple butterflies in the background" or "evenly scattered flower patterns". This repeated instance property introduces unique technical challenges for layout control in pattern generation compared with standard scene synthesis. First, text prompts struggle to precisely specify the number, spacing, and distribution of repeated elements. Second, many existing layout control methods assume that each region corresponds to a single large semantic object, making them not naturally suited to scenarios with many repeated small instances.

Existing controllable generation methods can be broadly divided into two categories. The first consists of training-based methods [1, 3, 10, 11, 17, 23, 24, 30], which introduce additional modules or conditional branches to encode control signals such as edges, depth, sketches, and layouts. While these methods often offer strong controllability, they typically require additional training data and substantial computational resources. The second category includes training-free inference-time control methods, which inject layout constraints during sampling through attention modulation, latent fusion, or optimization-based guidance. Compared with training-based approaches, they are more lightweight and flexible, yet they still face several limitations in pattern scenarios: some methods rely on iterative backward optimization [4, 5, 12, 26, 27], leading to high inference costs; others impose constraints through hard latent fusion or explicit mask-based stitching [2, 8, 15], which can easily introduce unnatural transitions at region boundaries; additionally, some methods intervene in internal attention through direct reassignment or weight adjustment during generation, which can lead to attention leakage, visual artifacts, and cluttered backgrounds.

Therefore, the core challenge we tackle is: *How can we achieve high-quality, controllable pattern generation without the overhead of training or the artifacts of existing inference-time methods?*

This paper introduces **PLAID** (**P**attern **L**ayout-**A**ware **I**nteractive **D**iffusion), a novel training-free method for layout-controllable pattern generation. Unlike approaches that directly impose layout constraints on latent variables or attention weights, the core of PLAID lies in foreground-background semantic decoupling within the cross-attention layers. During the early denoising stages, we keep the spatial query features from the image latents unchanged, while separately generating the corresponding key-value representations from the foreground prompt and the background prompt, respectively. These two branches are then continuously fused along the spatial dimension according to a layout mask. This design allows a soft layout guidance, avoiding the rigid shapes caused by hard boundary constraints, while also eliminating the need to construct fully separate branches or perform separate optimization for each repeated instance. In addition, we adopt point-based layouts as the minimal interaction unit, enabling users to control the spatial distribution of foreground elements at low interaction cost. Furthermore, since unnecessary structures often appear in background regions, we introduce a background color prior to the noise initialization stage to improve background purity. Extensive experiments verify the effectiveness and superiority of PLAID. In addition to pattern design, as shown in Fig. 1, our method can be applied to artistic text design and is compatible with general scene generation. Our method is not limited to two layers of foreground and background and is flexible to support multiple foreground branches.

In summary, our core contributions are as follows:

- We propose **PLAID**, a training-free layout control framework for pattern generation. PLAID uses a flexible dot-layout paradigm that supports both direct spatial placement and density-driven mask generation, enabling flexible control over both position and density at extremely low interaction cost.
- We introduce a **multi-branch attention decoupling paradigm** based on **Shared-Q and Split-KV**. By preserving the shared spatial query basis while separating text-conditioned branches and fusing them in the global refinement branch, PLAID provides layout guidance that is better aligned with pattern synthesis than hard latent fusion or optimization-based control.
- We introduce a **background purification mechanism** based on the initial distribution shift. By injecting a low-frequency background prior into the sampling initialization, this design biases early denoising toward cleaner and more appearance-consistent background configurations, improving foreground-background separation.

2 Related Work

Training-based Layout Control. Training-based methods [1, 3, 10, 11, 13, 17, 23, 24, 32] introduce additional parameters or adapters to encode spatial conditions. ControlNet [30] creates a trainable copy of the encoding layers to incorporate conditions like Canny edges or depth maps. SpaText [1] separates open-vocabulary scene control into spatio-textual representations, improving local editability. ControlNeXt [17] proposes a more efficient architecture to achieve powerful control with lower latency for both image and video generation. Additionally, AIComposer [10] focuses on compositional generation, utilizing feature integration to seamlessly combine arbitrary styles and content subjects. While these methods achieve impressive control, they typically require extensive training data and computational resources, limiting their flexibility for rapid prototyping in pattern design.

Training-free Layout Control. Training-free approaches [2, 4–6, 8, 12, 26, 27, 29] for layout control can be broadly categorized into two technical routes: *Backward Guidance* and *Forward Modulation*.

Backward Guidance methods treat layout control as an optimization problem. They compute losses based on attention maps and update the noisy latent via backpropagation to guide subsequent generation. For instance, Layout Guidance [4] systematically analyzes both forward and backward strategies and advocates the latter. BoxDiff [27] designs three dedicated losses (inner-box, outer-box, and corner losses) to iteratively optimize the latent. R&B [26] leverages differentiable minimum bounding rectangles and image edge cues (via Sobel operators) to resolve mismatches between layout constraints and actual object contours. CSG [12] computes multiple losses, such as inter-token competition constraints, to handle multi-semantic conflicts, while ZestGuide [5] requires precise object-shape masks and performs pixel-level adaptation based on cross-attention constraints. Although effective, these methods are computationally expensive because they require multiple backward passes at each denoising step. Moreover, methods such as BoxDiff and CSG rely on a one-to-one correspondence between tokens and boxes or masks, making them suitable for localizing distinct objects but less effective for generating *repeated instances* of the same semantic concept, which is crucial for pattern synthesis.

Forward Modulation methods avoid backpropagation by directly intervening in the forward inference process. DenseDiffusion [8] and Bounded Attention [6] modulate attention maps in a pretrained model according to textual and layout conditions by amplifying or suppressing cross-attention and self-attention values, which may lead to semantic leakage and cluttered backgrounds. Zero-Painter [15] further enables layout-conditioned synthesis through object masks, individual descriptions, and specially designed cross-attention mechanisms, but also requires strict object-shape masks. MultiDiffusion [2] runs multiple complete denoising branches and fuses them in the latent space according to masks. Although faster than optimization-based methods, such multi-branch approaches incur a high computational complexity, and latent-space fusion often results in hard composition, leading to unnatural boundaries. By comparison, our method enables effective soft attention fusion [28, 31] and introduces only a modest increase in computational cost.

Most existing layout control works focus on composite scenes with distinct objects rather than repeated foreground designs or pattern synthesis. Furthermore, nearly all spatial control methods rely on cumbersome bounding boxes or extremely dense text descriptions, revealing that user-friendly interaction paradigms remain largely unexplored. To reduce user interaction costs, we propose replacing traditional box constraints with a flexible dot-layout paradigm. Additionally, existing methods struggle to fundamentally suppress uncontrolled artifacts appearing in the background, whether through forward masking or backward loss calculation. We address this by intervening directly at the noise initialization stage to achieve Background Purification, which proves more effective than struggling with attention maps at every step.

3 PLAID Framework

This section introduces our PLAID framework, as illustrated in Fig. 2. First, the preliminaries on image synthesis based on latent diffusion models (LDMs) are reviewed in Sec. 3.1. We then introduce an overview of PLAID pipeline in Sec. 3.2. Then, we present our multi-branch pattern generation and background purification in Sec. 3.3 and Sec. 3.4.

3.1 Preliminaries

Our PLAID framework is built upon LDMs [21] with DDIM [22]. The forward process adds Gaussian noise to a clean image z_0 over T steps. The noisy image z_t at any step t is obtained as:

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \tag{1}$$

where $\bar{\alpha}_t$ is a pre-defined hyperparameter at step t, and $\epsilon \sim \mathcal{N}(0, \mathbf{I})$. To generate an image, we start from a random noise $z_T \sim \mathcal{N}(0, \mathbf{I})$ and iteratively perform denoising. Following DDIM [22], the image at the previous step z_{t-1} is estimated by first predicting the denoised image $\hat{z}_{t\to 0}$ and then blending it with the current noise estimate:

$$\hat{z}_{t\to 0} = (z_t - \sqrt{1-\bar{\alpha}_t}\epsilon_\theta(z_t, t, c))/\sqrt{\bar{\alpha}_t}, \tag{2}$$

$$z_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\hat{z}_{t\to 0} + \sqrt{1-\bar{\alpha}_{t-1}}\epsilon_\theta(z_t, t, c), \tag{3}$$

where $\epsilon_\theta(z_t, t, c)$ represents the noise predicted by the diffusion model based on the prompt c. z_0 is the final generated image. For simplicity, we omit the details regarding the latent space. The image is mapped into the latent space via a VAE.

3.2 Overview of PLAID Pipeline

The core idea of the PLAID framework is to utilize a minimalist spatial prompt to partition the image space into multiple object regions and a background region. By applying distinct textual conditions and denoising strategies to these specific regions during the diffusion inference process, we achieve controllable layout and density of pattern elements without the need for model fine-tuning.

As shown in Fig. 2, first, users provide and decompose the input prompt c_{full} into multiple object prompts $\{c_i\}_{i=1}^N$ and a background prompt c_{bg} (decomposition can be alternatively done by LLMs). Simultaneously, the spatial prompt, e.g., a set of layout masks $\{M_i\}_{i=1}^N$, is provided by the user or is randomly generated based on the preferred element density of the user. During the initial T steps of the denoising process, we decouple the generation into multiple foreground branches and a background branch. We employ a background purification strategy to optimize the background purity and utilize a layout-aware attention mechanism to fuse these branch features in the latent space based on the layout masks. In the later denoising stages, the process switches back to the standard global text prompt to unify the overall artistic style and refine details, yielding the final high-quality pattern.

Layout Generation. PLAID utilizes a set of lightweight gray-scale layout masks $\{M_i\}_{i=1}^N$, where each $M_i \in [0, 1]^{H\times W}$ serves as a spatial prior for the i-th object branch. Here, $M_i = 1$ indicates that a spatial location fully belongs to the i-th object region, while $M_i = 0$ indicates that it does not belong to that branch. Intermediate values between 0 and 1 indicate uncertain transition regions. During fusion, the gray value directly serves as a continuous mixing weight for the corresponding object attention. The background region is implicitly represented by the residual weight $M_{bg} = 1 - \sum_{i=1}^N M_i$. To ensure valid fusion weights, we constrain $\sum_{i=1}^N M_i \le 1$. We support three flexible layout modes with minimal interaction:

1. *Direct Masking:* Users directly draw dark dots or regions on a white canvas to indicate object locations. We normalize pixel intensities to $[0, 1]$ to form the corresponding mask(s).
2. *Precise Guidance:* When precision positioning is required, users can provide dot coordinates with a radius parameter. We convert each dot set into the corresponding mask.
3. *Density Guidance:* For users seeking quick generation without specific layout requirements, we synthesize layout masks from a target foreground coverage ratio ρ, the approximate size s of dots, a minimum inter-dot distance δ, and a maximum dot number K provided by users. We then generate the masks via stochastic placement: dots are sampled and iteratively added to the canvas while enforcing the minimum-separation constraint, until the estimated foreground coverage reaches ρ, producing layout masks with controllable density.

Prompt Decomposition. To achieve semantic decoupling among multiple object branches and the background, the original prompt c_{full} is decomposed into $\{c_i\}_{i=1}^N$ and c_{bg}. For example, if c_{full} = "white cloud and yellow sun in blue sky", then one possible decomposition is c_1 = "white cloud", c_2 = "yellow sun", and c_{bg} = "blue

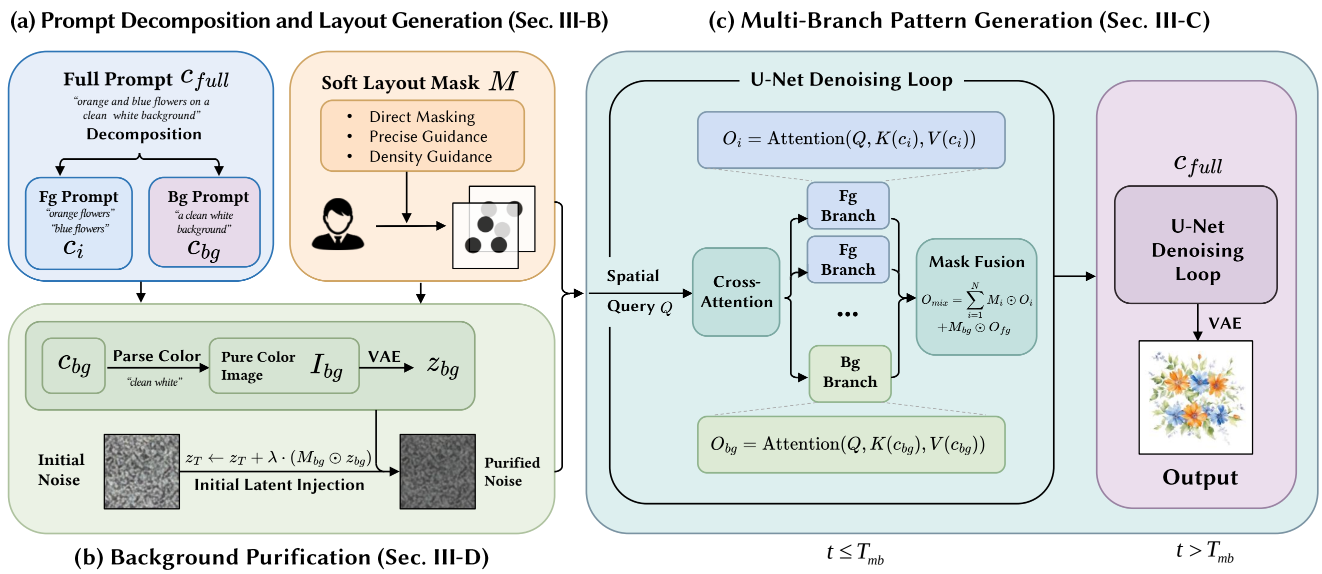


Figure 2: Illustration of the proposed PLAID framework.

sky". The decomposition can be performed manually by the user or automated via Large Language Models (LLMs).

3.3 Multi-branch Pattern Generation

In standard diffusion denoising steps, the Cross-Attention layers within the U-Net are responsible for aligning spatial features with textual semantics. For input spatial query features Q, the standard attention mechanism is calculated as follows:

$$\mathrm{Attention}(Q, K, V) = \mathrm{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V, \tag{4}$$

where the query feature Q is derived from the image latents with dimension d, while key K and value V are projected from the text embeddings.

Design Rationale. Our design is motivated by the observation that layout control in diffusion models should preserve the shared spatial basis of generation while reducing semantic interference across regions. Directly reweighting attention values is often sensitive because spatial cues are not carried only by a few content tokens, but are distributed across the prompt sequence and the attention computation. As a result, token-level amplification or suppression may provide unstable control and can easily introduce foreground leakage or background clutter in repeated-instance scenarios. Optimization-based latent updates are more flexible, but they require repeated backward passes during denoising and become less practical when many similar motifs must be arranged simultaneously. In our design, we share Q to preserve the shared spatial basis, and split KV to prevent information leakage without introducing high computational cost.

Layout-Aware Attention. We propose a Multi-branch Generation strategy incorporating Layout-Aware Attention. During the early denoising stage where $t \le T_{mb}$, we explicitly construct multiple sets of text embedding conditions based on the decomposed prompts $\{c_i\}_{i=1}^N$ and c_{bg}.

As illustrated in Fig. 2(c), in each cross-attention layer, we maintain Q unchanged to preserve shared spatial structural information, but separately compute the attention outputs for each object branch and the background branch:

$$O_i = \mathrm{Attention}(Q, K(c_i), V(c_i)), \quad i = 1, \ldots, N \tag{5}$$

$$O_{bg} = \mathrm{Attention}(Q, K(c_{bg}), V(c_{bg})). \tag{6}$$

Subsequently, we utilize the set of layout masks $\{M_i\}_{i=1}^N$ (downsampled to match the resolution of the current layer) to perform a spatial weighted fusion of these outputs, obtaining the final mixed attention output O_{mix}:

$$O_{mix} = \sum_{i=1}^{N} M_i \odot O_i + M_{bg} \odot O_{bg}. \tag{7}$$

Through this mechanism, the model is forced to attend to the corresponding object descriptions in regions covered by different object masks, while attending to the background description in the remaining regions. This spatial constraint effectively prevents semantic leakage across different object branches as well as between objects and the background, and inhibits background textures from interfering with the target layout.

When $t > T_{mb}$, we switch back to the standard denoising procedure conditioned on the full prompt c_{full}, which helps harmonize global style and suppress boundary inconsistencies introduced by early-stage region decoupling.

3.4 Background Purification

In pattern generation, the background is often expected to remain visually simple and semantically clean. However, diffusion models initialized from isotropic Gaussian noise may still allocate part of their generative capacity to unintended background structures, especially in regions that are only weakly constrained by text. To alleviate this issue, we introduce a background purification strategy at the *initialization stage*.

Our key idea is to perform an **initial distribution shift** by injecting a *low-frequency background prior* into the initial latent. This prior perturbs the initial condition of sampling so that the early denoising trajectory is biased toward visually cleaner and more appearance-consistent background configurations. Formally, given the layout, we modify the initial noise z_T as

$$z_T \leftarrow z_T + \lambda \cdot (M_{bg} \odot z_{bg}), \quad (8)$$

where z_{bg} denotes a latent background prototype and λ controls the strength of the shift.

In our implementation, we first parse the target background color from the prompt (or use a user-specified color) to construct a solid-color background image I_{bg}. We then encode it with the VAE encoder and average its latent over spatial dimensions to obtain a channel-wise background vector z_{bg}. Taking the spatial mean removes unnecessary spatial variation and retains only a coarse low-frequency appearance prior. Rather than enforcing hard constraints during denoising, this simple operation softly biases the generation process toward backgrounds with more consistent color and fewer unintended structures.

4 Experiments

4.1 Experimental Setup

Implementation Details. We implement PLAID on top of Stable Diffusion 1.5 (SD1.5) with DDIM sampling [22]. For fair comparison with competing methods, we instantiate PLAID with one foreground branch and one background branch. All experiments use $T = 50$ DDIM sampling steps and generate images at a resolution of 512×512. For multi-branch pattern generation, we set $T_{mb} = 25$. For the background purification module, we set the injection strength to $\lambda = 0.05$ by default, which provides a good empirical balance between visual quality and background cleanliness. We use GPT-4 as an auxiliary tool for foreground-background prompt decomposition. All experiments are conducted on a single NVIDIA RTX 4090 GPU.

Baselines. We compare PLAID with five representative methods that cover training-based control, training-free attention guidance, and training-free multi-path generation.

- **ControlNet** [30]. A representative *training-based* conditional control baseline. In our experiments, scribble maps are used as proxies for dot layouts for spatial guidance. ControlNet serves as a reference for hard condition control.
- **FreeControl** [14]. We include FreeControl as a general *training-free structural control* baseline. Different from methods designed for explicit box-, mask-, or point-based layout specification, FreeControl relies on reference-image-based structure guidance and appearance guidance, making it a useful reference for broader training-free controllable generation.
- **DenseDiffusion** [8]. We include DenseDiffusion as a representative *training-free forward attention modulation* baseline. It guides generation by modulating intermediate attention maps according to text and layout conditions.
- **R&B** [26]. We use R&B as a representative *training-free backward guidance* baseline. It enhances layout alignment during sampling through cross-attention guidance together with region-aware and boundary-aware objectives.
- **MultiDiffusion** [2]. We adopt MultiDiffusion as a representative *training-free multi-path generation* baseline. It binds and fuses multiple diffusion processes under shared constraints, and can incorporate spatial control signals without additional training.

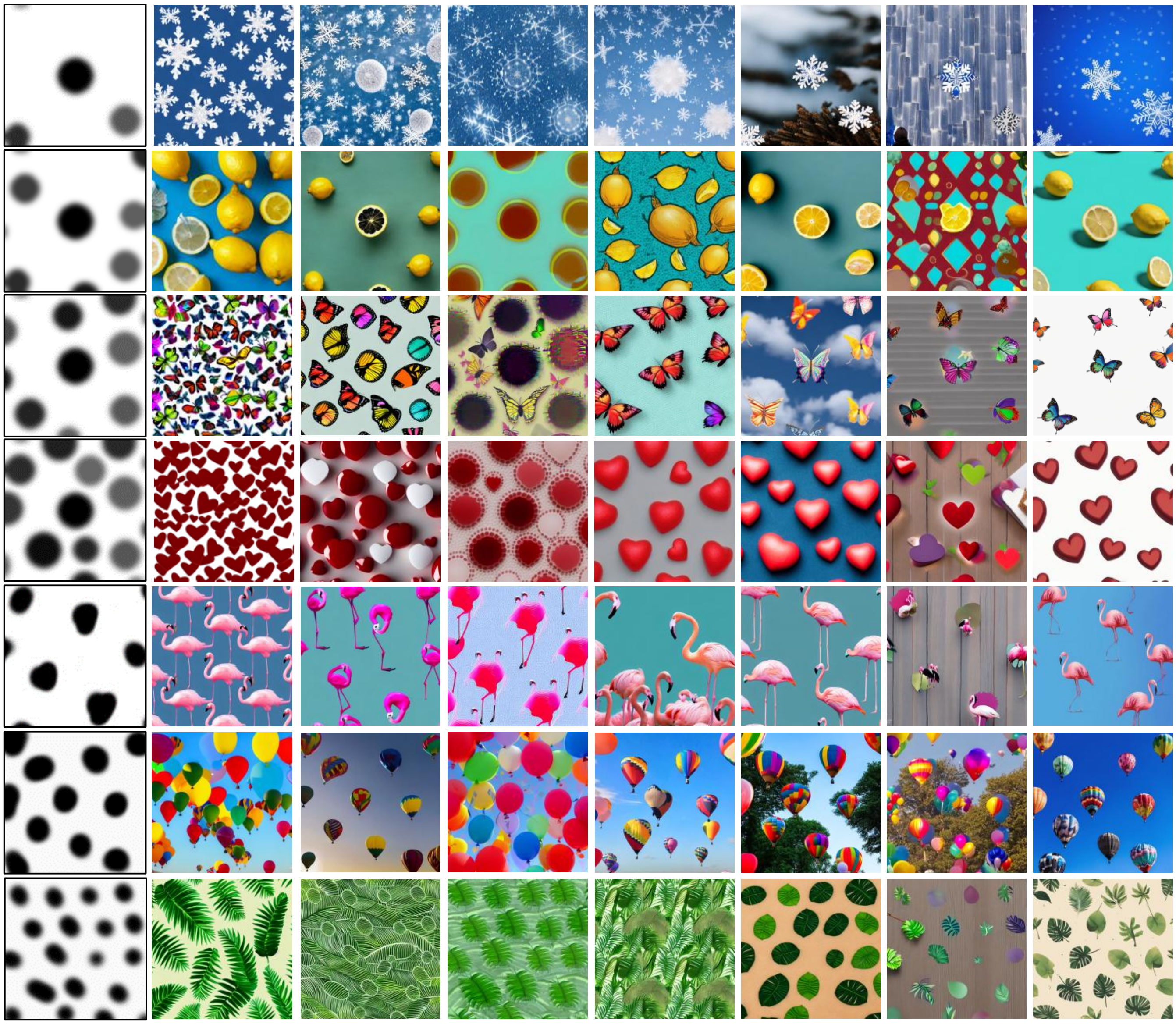
Evaluation Metrics. To comprehensively evaluate the generated results from the perspectives of semantic consistency, layout adherence, and visual quality, we adopt the following metrics:

- **Global CLIP** [19]: We compute the CLIP similarity between the generated image and the full text prompt to measure overall text-image semantic alignment.
- **Local CLIP** [19]: We evaluate local text-image consistency for foreground regions. Specifically, we first extract the foreground region from the generated image according to the mask, and then compute the CLIP similarity between this region and the foreground prompt, so as to more directly assess foreground semantic fidelity.
- **Centroid Distance**: To evaluate whether foreground elements appear at the desired locations, we use YOLOv11 [20] to detect foreground objects in the generated image and compute the Euclidean distance between the centroid of each detected foreground region and the centroid of the corresponding target mask region. A closer distance indicates better alignment between the generated foreground and the target layout.
- **Box IoU**: Based on YOLOv11 [20] detections, we further compute the Intersection-over-Union (IoU) [8] between the detected bounding boxes of foreground and the target layout regions, to measure the spatial overlap between foreground objects and target areas. A higher IoU indicates better layout adherence.
- **Aesthetic Predictor V2.5 and HPSv2** [25]: We use LAION Aesthetic Predictor V2.5 to evaluate overall aesthetic quality, and HPSv2 to measure alignment with human preferences. The former is a commonly used aesthetic scorer for generated images, while the latter is trained on large-scale human preference data and better reflects subjective human preference over generated results.

4.2 Quantitative Results

For quantitative evaluation, we tested on 40 diverse prompts with 8 different layout masks each (7 masks in Fig. 3 and 1 mask in Fig. 9), resulting in 320 prompt-mask combinations. As shown in Table 1, PLAID achieves the best overall performance in controllable pattern generation, topping the Global/Local CLIP, Centroid Distance, Box IoU, and Aesthetic V2.5 metrics. Notably, its lightweight, forward-only design requires only 1.68 seconds per image. This inference speed nearly matches the unconstrained SD baseline (1.57s) and significantly outperforms other controllable methods, especially optimization-heavy or preprocessing-heavy approaches like R&B (32.09s) and FreeControl (50.38s).

Prior methods struggle to balance semantic fidelity, layout adherence, and efficiency. While MultiDiffusion aligns layouts reasonably well, its latent-space fusion severely harms global coherence and aesthetics, as indicated by the worst scores on Aesthetic Predictor V2.5 and HPSv2. DenseDiffusion lacks explicit branch decoupling, frequently leading to foreground leakage and background clutter, obtaining worst Local CLIP. Moreover, ControlNet and FreeControl



(a) Layout (b) SD (c) ControlNet (d) FreeControl (e) DenseDiffusion (f) R&B (g) MultiDiffusion (h) PLAID (Ours)

Figure 3: Qualitative comparison on layout-controllable pattern generation. PLAID achieves a favorable balance between foreground quality, background cleanliness, and layout adherence. Prompts: "a lot of white snowflakes falling on a clean icy blue background"; "a lot of yellow lemons scattered on a solid turquoise background"; "a lot of colorful butterflies flying on a minimal white background"; "a lot of small red hearts repeating on a clean white background"; "a lot of pink flamingos standing closely on a simple pastel blue background"; "a lot of colorful balloons floating in a clear blue sky"; "a lot of green tropical leaves overlapping on a clean beige background".

tend to overfit the lightweight coarse masks used in our setting, causing distorted foreground structures, thus achieving low Local CLIP. R&B achieves strong layout and visual quality, but incurs slight foreground object distortions. In contrast, PLAID successfully avoids these drawbacks: our attention-level semantic decoupling ensures precise global and local alignment, while the soft spatial prior yields more flexible layout adherence without sacrificing visual naturalness or inference speed.

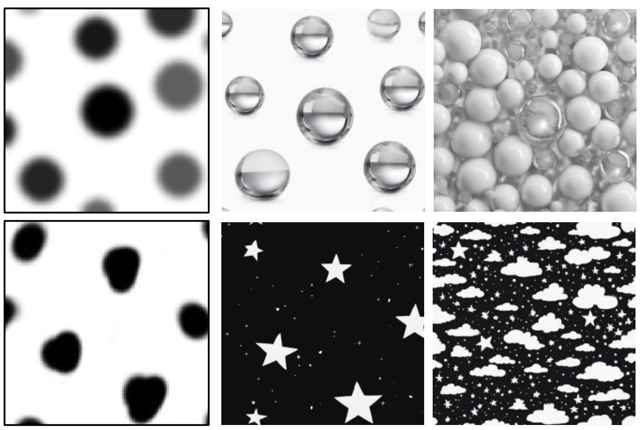
4.3 Qualitative Results

Figure 3 provides visual comparisons across all methods. Overall, compared with baselines, PLAID more consistently places repeated motifs in the intended regions while preserving natural object appearance and maintaining visually clean backgrounds.

By contrast, different baselines exhibit characteristic artifacts in our setting. ControlNet and FreeControl, which rely on stronger structural constraints, tend to overfit the coarse masks used in our

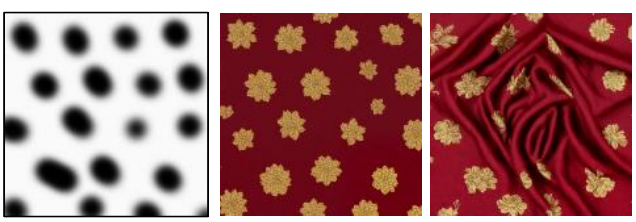
Table 1: Quantitative comparison on controllable pattern generation. PLAID achieves the best overall trade-off across semantic alignment (Global CLIP, Local CLIP), layout adherence (Centroid Distance, Box IoU), visual quality (Aesthetic V2.5, HPSv2), and efficiency. Runtime is reported as the average wall-clock time per image in our setting. For FreeControl, the runtime includes both PCA subspace construction and image generation. Bold and underline denote the best and second-best, respectively.

Method	Global CLIP (↑)	Local CLIP (↑)	Centroid Distance (↓)	Box IoU (↑)	Aesthetic V2.5 (↑)	HPSv2 (↑)	Time (s/img) (↓)
SD	0.2802	0.2234	58.5195	0.1102	5.4012	**0.2591**	**1.57**
ControlNet	0.2654	0.2247	41.5579	0.2382	5.0025	0.2318	3.24
FreeControl	0.2610	0.2201	47.3397	0.2577	4.9781	0.2219	50.38
DenseDiffusion	0.2522	0.2201	42.6927	0.2193	5.2019	0.2285	5.30
R&B	0.2397	0.2265	39.1967	0.3041	5.4068	0.2260	32.09
MultiDiffusion	0.2390	0.2312	36.1431	0.2783	4.3430	0.2162	3.73
PLAID (Ours)	**0.2832**	**0.2462**	**31.8047**	**0.3306**	**5.4243**	0.2488	1.68



(a) Layout (b) Full (c) w/o BG Purification

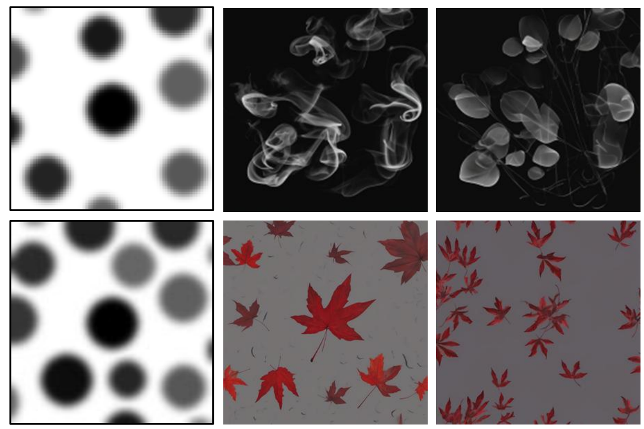
Figure 4: Ablation study on background purification. Prompts: "shiny crystal spheres floating against a pure white studio background"; "a dark indigo night sky scattered with bright stars".



(a) Layout (b) High λ: Purer BG (c) Low λ: Complex BG

Figure 5: Effect of λ in balancing purity and diversity. Prompt: "golden floral patterns decorating a dark crimson red fabric background".

task, often leading to distorted or overly rigid foreground shapes (rounded butterflies and hearts). DenseDiffusion sometimes shows foreground leakage (leaves case) or cluttered backgrounds (lemon case), suggesting incomplete foreground-background separation. R&B generally provides reasonable layout alignment, but its results can still appear less clean in background (snowflake and balloon cases). MultiDiffusion tends to introduce greyish results with visible seams or inconsistent transitions due to latent-space fusion. In



(a) Layout (b) Full (c) w/o c_{full} Refinement

Figure 6: Ablation study on full prompt refinement. Prompts: "thin white smoke wisps drifting across a dark black background"; "bright red autumn leaves floating in a dark gray sky".

comparison, PLAID benefits from attention-level semantic decoupling, soft region-wise fusion, and background purification, yielding cleaner compositions and more harmonious foreground repetition across a wide range of pattern types.

4.4 Ablation Study

Effect of Background Purification. We compare the full model ($\lambda = 0.05$) with a variant disabling latent injection ($\lambda = 0$) in Fig. 4. Without background purification, the background region tends to be more cluttered, which interferes with the overall visual cleanliness of the generated pattern. Background purification provides a stable low-frequency prior that steers generation toward cleaner backgrounds while preserving foreground semantic fidelity.
Parameter Sensitivity of λ. We evaluate different injection strengths in Fig. 5. For cases where more complex backgrounds are desired, a smaller λ can be used. In contrast, larger values tend to increase background purity. We find $\lambda = 0.05$ to be an empirical choice that balances background purity and diversity.
Effect of Global Refinement. We compare the default setting with a variant that keeps multi-branch mixing throughout the entire denoising process in Fig. 6. Maintaining multi-branch mixing

(a) Layout (b) SD1.5 (c) SD2.1 (d) SDXL

Figure 7: PLAID's plug-and-play nature on different backbones. Prompt: "bright red autumn leaves floating in a dark gray sky".

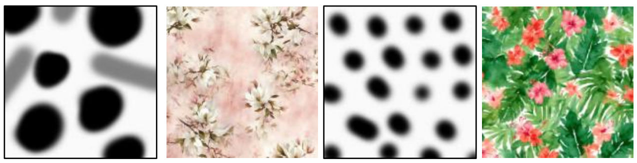


Figure 8: Pattern design with professional textile LoRA.

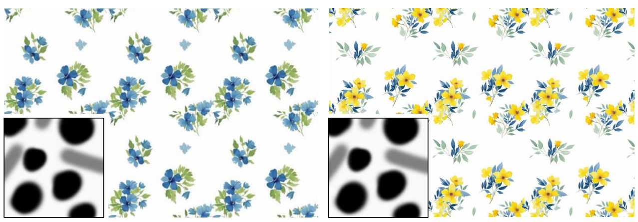


Figure 9: Periodic tiling pattern design.

all the way to the end may introduce discontinuities near region boundaries. Switching back to the full prompt in the final 50% of the denoising steps improves global stylistic consistency and yields more natural and coherent results.

4.5 More Results

Transferability of the Base Model. Our training-free PLAID can be seamlessly plugged into text-to-image diffusion models. In Fig. 7, we successfully applied PLAID to SD1.5, SD2.1 and SDXL. Its plug-and-play nature makes it well-positioned to benefit from future base model improvements.

Practical Compatibility. We tested PLAID on an SDXL LoRA trained on professional textile printing datasets (sourced from yibaiaigc.com). It is evident that with the integration of LoRA, our method produces pattern designs that feature richer layering and exhibit a more professional aesthetic.

In practice, dyeing and painting are typically performed on large sections of various types of large surfaces, thus patterns should consist of repetitive splicing of periodic units. PLAID is compatible with this practical design workflow. We applied PLAID to the Seamless Tiling model [9] and Fig. 9 shows that PLAID's shape-flexible nature maintains tileability even with non-periodic dot layouts.

Applications in General Object Synthesis. PLAID is already capable of generating general scenarios such as "a cat on a chair in a white room" in Fig. 10. Furthermore, we can divide the foreground objects into different groups based on their color or semantics, and control their layouts separately, as illustrated in Fig. 2 and Fig. 11.

Robustness to Layout Density. In addition to the lemon examples in Fig. 1, we provide additional results with various layout density in Fig. 12. Across a wide range of densities, PLAID shows consistent

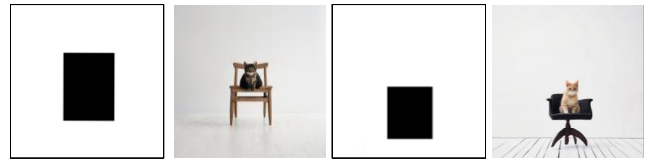
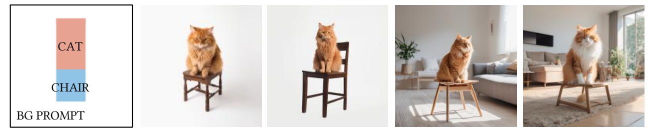


Figure 10: Performance on general object synthesis.



BG PROMPT: a white room BG PROMPT: a bright living room

Figure 11: Performance on general object synthesis with two foreground object masks.

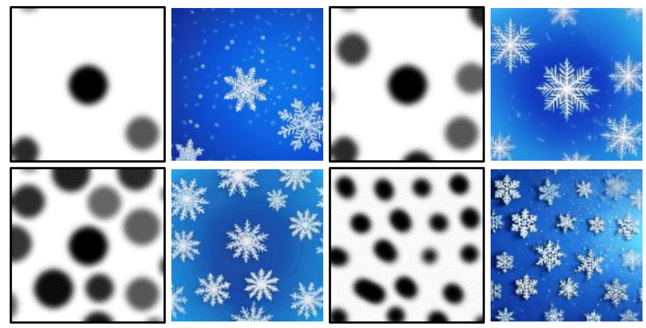


Figure 12: Results of different densities.

performance. Since our method avoids iterative per-instance optimization and performs foreground-background decoupling through shared-query attention fusion, it remains well suited to repeated-instance layouts even when motif density becomes high. This robustness is important for pattern design, where users may wish to vary density substantially without redesigning the control interface.

5 Conclusion

We present **PLAID**, a training-free framework for layout-controllable pattern generation. By introducing an extremely flexible dot-based interaction paradigm, PLAID significantly reduces user effort while supporting flexible density guidance. A multi-branch diffusion process is designed to disentangle foreground and background generation through an attention-level Shared-Q, Split-KV fusion. Coupled with a background purification mechanism, our method robustly prevents semantic leakage and background hallucinations. Extensive evaluations demonstrate that PLAID achieves state-of-the-art layout adherence and visual quality. PLAID also shows promising compatibility with artistic text design, general object synthesis, and diverse diffusion backbones. We hope that PLAID can serve as a useful step toward lightweight and user-friendly generative tools in artistic design and beyond. While PLAID is effective in most cases, it can occasionally introduce minor artifacts or discontinuities near region boundaries, especially for patterns requiring strong long-range structural consistency. Future work may explore extending the interactive prompts beyond dots and adapting the framework for dynamic or 3D pattern synthesis.

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